Endocrine-disrupting chemicals and obesity development in humans: a review.

. . . . .

This study reviewed the literature on the relations between exposure to chemicals with endocrine-disrupting abilities and obesity in humans. The studies generally indicated that exposure to some of the endocrine-disrupting chemicals was associated with an increase in body size in humans. The results depended on the type of chemical, exposure level, timing of exposure and gender. Nearly all the studies investigating dichlorodiphenyldichloroethylene (DDE) found that exposure was associated with an increase in body size, whereas the results of the studies investigating polychlorinated biphenyl (PCB) exposure were depending on dose, timing and gender. Hexachlorobenzene, polybrominated biphenyls, beta-hexachlorocyclohexane, oxychlordane and phthalates were likewise generally associated with an increase in body size. Studies investigating polychlorinated dibenzodioxins and polychlorinated dibenzofurans found either associations with weight gain or an increase in waist circumference, or no association. The one study investigating relations with bisphenol A found no association. Studies investigating prenatal exposure indicated that exposure in utero may cause permanent physiological changes predisposing to later weight gain. The study findings suggest that some endocrine disruptors may play a role for the development of the obesity epidemic, in addition to the more commonly perceived putative contributors.